
DDA5001 MACHINE LEARNING FINAL PROJECT

PART III: TEST-TIME SCALING ON QWEN MODEL

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ABSTRACT

We evaluate the mathematical reasoning performance of the base and GRPO-tuned instruction-finetuned variants of Qwen2.5-Math-1.5B across three benchmarks—Math500, AMC23, and AIME25—under varying sampling temperatures ($\tau = 0.6, 1.0, 1.2$) and increasing numbers of samples (pass@k). Our results show that higher temperature consistently degrades performance, especially with smaller k , while larger k mitigates this drop by improving solution coverage. The instruction-tuned model significantly outperforms the base model across all settings, highlighting the effectiveness of instruction tuning in enhancing both accuracy and robustness in mathematical reasoning.

1 Results

Table 1: Base model (Qwen2.5-Math-1.5B) performance across datasets and temperatures. All values in %.

Dataset	Temp	pass@1	pass@2	pass@4	pass@8	pass@16	pass@32	pass@64	maj@1
Math500	0.6	35.91	51.35	65.83	76.86	84.60	—	—	44.40
	1.0	29.12	45.17	62.16	76.25	85.60	—	—	43.00
	1.2	17.84	30.60	47.47	65.24	79.20	—	—	16.20
AMC23	0.6	34.77	48.36	60.74	70.93	79.26	85.75	90.00	47.50
	1.0	24.57	38.54	53.71	67.60	79.00	86.51	90.00	42.50
	1.2	14.41	25.11	39.85	56.29	71.12	82.67	90.00	7.50
AIME25	0.6	4.11	7.16	11.34	16.19	21.29	27.42	36.67	6.67
	1.0	2.71	5.00	8.63	13.53	19.34	26.60	36.67	3.33
	1.2	1.77	3.39	6.25	10.78	16.83	23.47	30.00	0.00

*Coding.

†Report writing.

¹Training.

Table 2: Instruct model (Qwen2.5-Math-1.5B-Instruct) performance across datasets and temperatures. All values in %.

Dataset	Temp	pass@1	pass@2	pass@4	pass@8	pass@16	pass@32	pass@64	maj@1
Math500	0.6	75.11	80.86	85.54	89.01	91.00	–	–	79.20
	1.0	74.60	80.88	85.39	88.86	91.60	–	–	79.80
	1.2	70.14	77.98	84.00	88.60	91.80	–	–	76.80
AMC23	0.6	53.52	63.85	72.47	79.22	84.85	89.87	95.00	60.00
	1.0	52.07	63.05	71.35	77.67	83.69	89.55	95.00	65.00
	1.2	48.28	59.44	69.03	76.70	82.32	86.03	90.00	62.50
AIME25	0.6	8.39	13.76	20.12	26.70	33.80	41.52	50.00	13.33
	1.0	9.48	15.27	22.03	29.16	36.79	44.20	50.00	16.67
	1.2	7.29	12.13	18.02	23.99	29.96	36.42	43.33	16.67

Table 3: Performance gain of Instruct model over Base model (Qwen2.5-Math-1.5B-Instruct minus Qwen2.5-Math-1.5B). All values in %.

Dataset	Temp	pass@1	pass@2	pass@4	pass@8	pass@16	pass@32	pass@64	maj@1
Math500	0.6	39.20	29.51	19.71	12.15	7.00	–	–	34.80
	1.0	45.48	35.71	23.23	12.61	6.00	–	–	36.80
	1.2	52.30	47.38	36.53	23.36	12.60	–	–	60.60
AMC23	0.6	18.75	15.49	11.73	8.29	5.59	4.12	5.00	12.50
	1.0	27.50	24.51	17.64	10.07	4.69	3.04	5.00	22.50
	1.2	33.87	34.33	29.18	20.41	11.20	3.36	0.00	55.00
AIME25	0.6	4.28	6.60	8.78	10.51	12.51	14.10	13.33	6.66
	1.0	6.77	10.27	13.40	15.63	17.45	17.60	13.33	13.34
	1.2	5.52	8.74	11.77	13.21	13.13	12.95	13.33	16.67

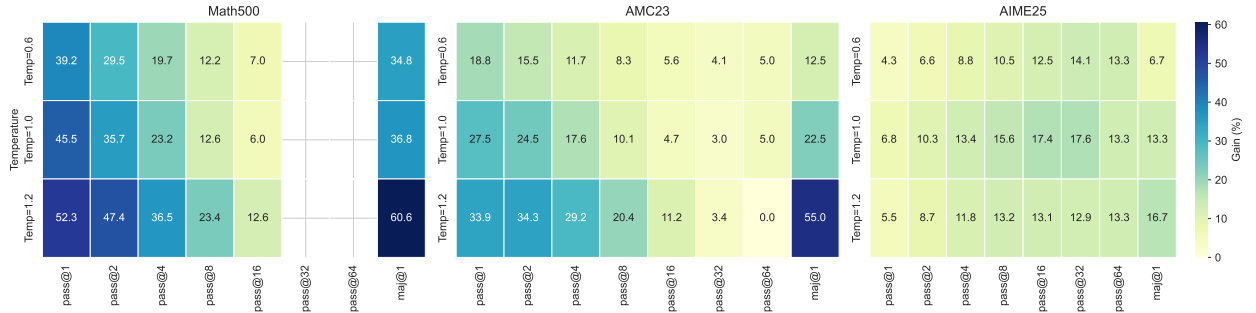


Figure 1: Heat map of performance gain of Instruct model over Base model.

2 Observations and Analysis

2.1 Performance Degradation with Higher Temperature

Raising the sampling temperature consistently degrades performance across nearly all models and test sets, as evidenced in Tables 1 and 2. The drop is most pronounced for small k (e.g., `pass@1` and `pass@2`), where accuracy declines sharply as temperature increases from 0.6 to 1.2. In contrast, metrics with larger k (e.g., `pass@32`, `pass@64`) are relatively more stable, suggesting that higher-temperature sampling preserves some correct solutions within a broader set of diverse outputs—though single-sample reliability suffers significantly.

2.2 Improvement in `pass@k` as k Increases

As k increases, `pass@k` consistently improves across all datasets and models, as shown in Tables 1 and 2, reflecting the benefit of multiple sampling: with more attempts, the likelihood of generating at least one correct solution rises. The gain is most rapid from `pass@1` to `pass@8`, especially on moderately difficult benchmarks like **AMC23**, where performance often jumps by 30–40 percentage points. On harder tasks such as **AIME25**, improvements continue steadily up to `pass@64`, though absolute scores remain low, indicating that even extensive sampling cannot fully compensate for fundamental reasoning gaps. Overall, larger k reliably boosts success rates, but with diminishing returns beyond `pass@16` in many cases.

2.3 Enhanced Accuracy and Robustness via Instruction Tuning

The performance gain of the GRPO-tuned Instruct model over the Base model (Qwen2.5-Math-1.5B-Instruct minus Qwen2.5-Math-1.5B) is consistently positive across all datasets and evaluation metrics, as shown in Table 3 and Figure 1. The gains are most pronounced on **Math500**, where improvements exceed 50 percentage points in `maj@1` at $\tau = 1.2$, and reach up to 52.3% in `pass@1`. This indicates that instruction tuning significantly enhances both single-sample accuracy and consensus-based reasoning under high-diversity sampling.

On **AMC23**, the Instruct model achieves substantial gains—up to 34.33% in `pass@2` at $\tau = 1.2$ —and maintains strong performance even at high temperatures, with a notable 55.00% improvement in `maj@1`. The heat map reveals that these gains are relatively stable across k , suggesting improved consistency in reasoning paths.

For the hardest benchmark, **AIME25**, the performance boost is more modest but still significant, especially at higher k . The gain in `pass@16` reaches 17.45% at $\tau = 1.0$, and remains above 12% for all $k \geq 8$. Importantly, the Instruct model shows increasing gains with larger k , particularly at $\tau = 1.0$, indicating better exploration of correct reasoning trajectories.

Overall, the heat maps confirm that instruction tuning not only improves absolute accuracy but also enhances robustness to temperature variation, especially in majority-voting scenarios. While the Base model’s performance degrades sharply at $\tau = 1.2$ (e.g., `maj@1` drops to 7.5% on **AMC23**), the Instruct model maintains high consensus, demonstrating superior generalization and coherence under stochastic decoding.

3 Difficulties

The main difficulty lay in the initial environment configuration, particularly with the `vLLM` library, which exhibited instability across different system setups. Once resolved, all subsequent steps were executed without substantial issues.